

```

3. apple.loc[apple['SMA20'] > apple['SMA50'], 'SMA_Cross'] = 1 # Bullish
4. apple.loc[apple['SMA20'] < apple['SMA50'], 'SMA_Cross'] = -1 # Bearish
5.
6. # Identify MACD crossover signals
7. apple['MACD_Cross'] = 0 # Initialize with neutral signal
8. apple.loc[apple['MACD'] > apple['MACD_Signal'], 'MACD_Cross'] = 1 # Bullish
9. apple.loc[apple['MACD'] < apple['MACD_Signal'], 'MACD_Cross'] = -1 # Bearish
10.

```

Common crossover signals include:

- **Moving Average Crossovers:** A shorter-term MA crossing above a longer-term MA (bullish) or below it (bearish)
- **MACD Line and Signal Line:** The MACD line crossing above the signal line (bullish) or below it (bearish)
- **Price and Moving Average:** Price crossing above a moving average (bullish) or below it (bearish)
- **Stochastic %K and %D:** The %K line crossing above the %D line (bullish) or below it (bearish)

While crossovers provide clear, objective signals, they can generate false signals in choppy, sideways markets. Many algorithmic systems use additional filters to reduce these false signals.

Divergence and Convergence: When Price and Indicators Disagree

Divergence occurs when the price movement and an indicator trend in opposite directions, potentially signaling a weakening trend and upcoming reversal.

```

1. def detect_divergence(price, indicator, window=10):
2.     """
3.     Detect potential divergence between price and an indicator

```